Alisina Bayati

Ph.D. Candidate — Optimization and Control

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Citizenship: United States | Iran

Education

University of Illinois at Urbana–Champaign *2020– 2026 (expected)*
Ph.D. in Mechanical Science & Engineering
Advisor: Prof. Srinivasa Salapaka

University of Illinois at Urbana–Champaign *2023 - 2025*
M.S. in Mathematics GPA 3.90/4.00

Sharif University of Technology *2015– 2019*
B.S. in Mechanical Engineering GPA 18.10/20 (3.89/4.00)
Advisor: Prof. Hamed Moradi

Publications

Peer-Reviewed

[1] **A. Bayati**†, D. Tiwary†, S. Salapaka. "A Control Barrier Function Approach to Constrained Resource Allocation Problems in a Maximum Entropy Principle Framework." *IEEE CDC 2025.* [IEEE Xplore], Code

[2] S. Basiri, **A. Bayati**, S. Salapaka. "Orthogonal non-negative matrix factorization with sparsity constraints." *IEEE CDC 2025.* [IEEE Xplore], Code

[3] **A. Bayati**, A. Srivastava, V. Mundada, S. Salapaka, H. Feng, A. Malvandi. "Energy efficiency in industrial drying: A hybrid ultrasonic system with a novel dynamic optimization framework" *Energy Conversion & Management.* (in press) [DOI]

[4] A. Srivastava†, **A. Bayati**†. "Sparse linear regression with constraints: A flexible entropy-based framework." *IEEE ECC 2024.* [IEEE Xplore]

[5] **A. Bayati**, A. Srivastava, A. Malvandi, H. Feng, S. Salapaka. "Towards efficient modularity in industrial drying: A combinatorial optimization viewpoint." *IEEE ACC 2023.* [IEEE Xplore]

In Preparation

[6] **A. Bayati**, A. Srivastava, S. Salapaka. "Dynamic Resource Allocation under Safety and Mobility Constraints: A Maximum Entropy Framework for Multi-Agent Systems."

Experience & Projects

Graduate Research Assistant, SENSIC Lab, UIUC *2020 – Present*

- **Control-Theoretic Approaches to Combinatorial Optimization** *2024 – Present*
  - Designed a dynamic feedback strategy for general constrained optimization problems that provably drives decision variables toward KKT stationarity, with guarantees on asymptotic convergence and feasibility.
  - Applied to capacitated facility location problems, achieving up to 20× and 240× speedups over Safe Gradient Flow (Cortés et al.) and `SciPy`'s SLSQP. Paper Code
  - Also applied to orthogonal non-negative matrix factorization with feature sparsity. Paper Code
  - Ongoing work extends to dynamic settings, where the objective and constraints evolve over time (e.g., collision-avoidance constraints in multi-agent systems), and the goal is to continuously track the moving optimizer in real time.
- **Energy-Efficient Optimization for Hybrid Drying Systems** *2020 – Present*
  - Developed a mixed-integer, multi-stage optimization framework for ultrasonic/convective drying, reducing energy consumption by 13% relative to the static single-stage baseline while satisfying key product quality constraints (e.g., color, moisture). Results under review at the *Journal of Energy Conversion and Management.*
  - Currently extending to a smart drying setup involving multiple drying technologies, jointly optimizing the sequence of operations and their conditions.

- **Routing Optimization in Residential Wi-Fi Mesh Networks** *2021 – 2023*
  - Built a real-time digital twin for Wi-Fi mesh networks using permutation-invariant DNNs; solved large-scale placement and routing problems.
  - Achieved latency in the lowest 3rd percentile across 50,000 uniformly sampled brute-force configurations.
  - *Funded by Foxconn Interconnect Technology (FIT).*

Graduate Teaching Assistant, University of Illinois (UIUC) *2023 – Present*

- ME 340: Dynamics of Mechanical Systems
- TAM 541: Mathematical Methods I (Graduate Level)
- TAM 210/211: Statics & Intro to Dynamics
  - Ranked "*Excellent Teaching Assistant*" in campus-wide student evaluations.

Undergraduate Researcher, Sharif University of Technology *2018 – 2019*

- **Optimal Control of Drug Delivery in Nonlinear Cancer Models**
  - Designed optimal PID controllers for nonlinear cancer-tumor drug-delivery models in MATLAB/Simulink, minimizing side effects while reaching the target tumor volume.
  - Conducted as part of the Process Control Lab under Prof. Hamed Moradi

Selected Graduate Coursework

Mathematics
- Real Analysis
- Functional Analysis
- Random Processes
- Vector-Space Optimization
- Abstract Linear Algebra
- Abstract Algebra

Machine Learning & Statistics
- Statistical RL
- Machine Learning
- Computer Vision
- Statistics & Probability

Control & Systems
- Optimal Control Theory
- Nonlinear Systems
- Linear Control Systems
- Dynamic Programming

Honors & Awards

- **CDC Student Travel Award**, IEEE Conference on Decision and Control *2025*
- **Teachers Ranked as Excellent**, University of Illinois Urbana–Champaign *Spring 2024*
- **ACC Student Travel Award**, American Control Conference *2023*
- **Graduate Admission Fellowship (One-Year)**, MechSE Department, UIUC *2020–2021*
- **Top 0.2% National Rank** among 182,000 participants in Iran's Nationwide University Entrance Exam (Math & Physics) *2015*

Technical Skills

Programming	Python (NumPy, pandas, PyTorch, SciPy, CVXPY, scikit-learn), MATLAB, C, LaTeX
Tools	Jupyter, Simulink, OpenCV
Languages	Persian (native), English (fluent), Arabic (elementary)

Service

- Peer Reviewer, ACC, ECC, CDC, L4DC *2023–Present*

References

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Additional references available upon request.